

Effect of Low Moderate and High Intensity Plyometric Training on Leg Strength and Kicking Performance of Football Players

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ABSTRACT

The purpose of the study was to find out the effect of low, moderate and high intensity plyometric training on leg strength and kicking performance of Football players. To achieve this purpose, sixty male Football players were selected as subjects, randomly, from Annamalai University, Chidambaram. The selected subjects were divided into four equal groups of fifteen subjects each, namely low, moderate and high intensity plyometric training groups and the control group. Experimental groups were trained for three alternative days, in a week, for twelve weeks, with 50 – 65%, 65 – 80% and 80 – 95% of intensity, for low, moderate and high intensity plyometric training. Leg strength and kicking were selected as criterion variables, and they were tested prior to and immediately after the training programme, by using leg lift with dynamometer and Sir Bobby Charlton Soccer School of Australia Skill Test for kicking, respectively. The ANCOVA was applied to find out the significant difference in each criterion variables, if any, among the groups. The Scheffe's Post Hoc Test was applied to determine the paired mean differences, if any.

The results of the study revealed that there was a significant improvement on leg strength and kicking performance, of the training groups, due to low, moderate and high intensity plyometric training, as compared to the control group.

INTRODUCTION

Now-a-days, all Football players are facing some unique challenges to develop the required motor fitness level, to execute the skills while playing in game situation in Football. Motor fitness status of each individual is directly related

to the skill performance of the Soccer. Strength is key to success in modern athletics. It is much easier to improve than the techniques. Strength may be defined as the neuromuscular capacity to overcome an external and internal resistance. The skill performance of Football players fully

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depends on the motor fitness components; though there are several training methods which are recommended for the improvement of skill performance. Plyometric training can take many forms, including jump training for the lower extremities and medicine ball exercises for the upper extremities. All the exercises are progressive in nature, with a range of low to high intensity in each type of exercises. In plyometrics, intensity is controlled by the type of exercise performed.

Hypotheses

The purpose of the study was to find out the effect of low, moderate and high intensity plyometric training on leg strength and kicking performance of Football players. The investigation framed the following hypotheses: there may be significant improvement on leg strength and kicking performance due to twelve weeks of low, moderate and high intensity

plyometric training, as compared to control group; and also, there may be a significant difference on leg strength and kicking among low, moderate and high intensity plyometric training groups.

METHODOLOGY

To achieve this purpose of the study, sixty male Football players from Annamalai University, Chidambaram, were selected as subjects, at random. The age of the subjects ranged from 18 to 25 years. The selected subjects were divided into four equal groups of fifteen subjects each. Group I underwent low intensity plyometric training (50-65%), Group II underwent moderate intensity plyometric training (65-80%), Group III underwent high intensity plyometric training (80-95%) and Group IV acted as the control group.

Load Dynamics

The intensity variation in 12 weeks training for low, moderate and high intensity plyometric training groups are given in Table I.

Table-1 : Per centage of Intensity of Plyometric Training of the three Experimental Groups.

Groups Weeks	Percentage of Intensity		
	Low Intensity Plyometric Training Group (%)	Moderate Intensity Plyometric Training Group (%)	High Intensity Plyometric Training Group (%)
1-3	50	65	80
4-6	55	70	85
7-9	60	75	90
10-12	65	80	95

(5% of intensity was increased at once in three weeks)

Assessment of Workload for Plyometric Training

- a. The exercise when performed against time, the time taken was noted. Later the same exercise was performed under extended time.

Example: *Single leg Zig-Zag Hop*,

Distance = 10 metres.

Individual's best performance = 10 secs.

(100% Capacity)

60% = Performing the same in 14 secs.

65% = Performing the same in 13.5 secs.

- b. When the consecutive exercise were performed like two-foot ankle hop/split squat jump/pull-over pass, the number of times the same act was performed continuously in a given time was noted. Later, the same exercise was performed with less number of two-foot ankle hop/ repetitions, keeping time constant.

Example: *Pull-over pass*,

Time: 1 minute

Individual's best performance = 20 counts

(100% capacity)

60% Capacity = 12 counts

65% Capacity = 13 counts

Plyometric Exercises for Soccer Players

- | | | |
|----------------|---|-------------------------------|
| Jumping place | - | 1. Two foot ankle hop, |
| | | 2. Split squat jump |
| Standing jump | - | 1. Lateral jump over barriers |
| Multiple jumps | - | 1. Hexagon drill, |
| | | 2. Cone hops with 180° turn |

- | | | |
|-------------------------|---|--------------------------|
| Box drill | - | 1. Alternating push off |
| Depth jump | - | 1. Jump to box, |
| | | 2. Single leg depth jump |
| Bounding | - | 1. Power skipping, |
| | | 2. Single leg bounding |
| Medicine ball exercises | - | 1. Heel toss, |
| | | 2. Pull over pass |

Leg strength and kicking were selected as criterion variables and assessed by using leg lift with dynamometer and Sir Bobby Charlton Soccer School of Australia skill test for kicking, respectively. The experimental design, was used for the present study, was random group design. The analysis of covariance used to analyse the significant difference in each criterion variables among the groups. Since, four groups were compared, whenever the obtained 'F' ratio for adjusted post-test means was found to be significant, the Scheffe's test to find out the paired mean differences, if any, were used. The level of significance to test the 'F' ratio obtained by the analysis of covariance was 0.05, which was considered appropriate.

RESULTS & DISCUSSION

The data collected prior to and after the experimental period on leg strength and kicking, of low intensity plyometric training group, moderate intensity plyometric training group, high intensity plyometric training group and control group, were analysed and presented in Table 1.

Table-1 : Analysis of Covariance of Data on Leg Strength and Kicking Between Pre and Post Test of Low Intensity Plyometric Training Group, Moderate Intensity Plyometric Training Group, High Intensity Plyometric Training Group and Control Group

	Test	Low Intensity Plyometric Training Group	Moderate Intensity Plyometric Training Group	High Intensity Plyometric Training Group	Control Group	Source of Variances	Sum of Squares	df	Mean Squares	Obtained 'F' Ratio
Leg Strength	Pre-test									
	Mean	92.54	92.27	92.12	92.46	Between	1.61	3	0.54	2.25
	SD	0.55	0.43	0.46	0.50	Within	13.21	56	0.24	
	Post-test									
	Mean	95.66	97.26	98.41	92.46	Between	300.45	3	100.15	182.09*
	SD	0.63	0.93	0.87	0.39	Within	30.60	56	0.55	
Kicking	Adjusted Post-test									
	Mean	95.66	97.26	98.41	92.46	Between	282.30	3	94.10	168.04*
						Within	30.60	55	0.56	
	Pre-test									
	Mean	36.43	36.13	36.48	36.61	Between	1.87	3	0.62	2.07
	SD	0.62	0.50	0.50	0.54	Within	16.54	56	0.30	
Kicking	Post-test									
	Mean	41.84	44.50	47.51	36.70	Between	946.40	3	315.47	157.74*
	SD	2.00	1.57	1.04	0.70	Within	112.21	56	2.00	
	Adjusted Post-test									
	Mean	41.84	44.50	47.51	36.70	Between	922.24	3	307.41	150.69*
						Within	112.21	55	2.04	

The table value required for significance at .05 level with df 3 and 56 & 3 and 55 are 2.772 and 2.775, respectively.

* Significant at .05 level of confidence.

Table 1 shows that the obtained "F" ratio value 168.04 for adjusted post test mean values on leg strength which was greater than the required table value of 2.775 for significance with df 3 and 55. The results of the study showed that there was a significant difference among four groups, on leg strength.

Table 1 also shows that the obtained "F" ratio value 150.69 for adjusted post test mean values on kicking which was greater than the required table value of 2.775 for significance with df 3 and 55. The results of the study showed that there was a significant difference among four groups, on kicking.

Table-2 : Scheffe's Post Hoc Test for the Difference Between Six Paired Adjusted Post Test Means of Leg Strength and kicking

	Adjusted Post-test Means				Mean Difference	Confidence Interval
	Low Intensity Plyometric Training Group	Moderate Intensity Plyometric Training Group	High Intensity Plyometric Training Group	Control Group		
Leg Strength	95.66	97.26	—	—	1.60*	0.788
	95.66	—	98.41	—	2.75*	0.788
	95.66	—	—	92.46	3.20*	0.788
	—	97.26	98.41	—	1.15*	0.788
	—	97.26	—	92.46	4.80*	0.788
	—	—	98.41	92.46	5.95*	0.788
Kicking	41.84	44.50	—	—	2.66*	1.51
	41.84	—	47.51	—	5.67*	1.51
	41.84	—	—	36.70	5.14*	1.51
	—	44.50	47.51	—	3.01*	1.51
	—	44.50	—	36.70	7.80*	1.51
	—	—	47.51	36.70	10.81*	1.51

*Significant at 0.05 level of confidence.

Table 2 shows that the adjusted post test mean difference between low intensity plyometric training group and moderate intensity plyometric training group, low intensity plyometric training group and high intensity plyometric training group, low intensity plyometric training group and control group, moderate intensity plyometric training group and high intensity plyometric training group, moderate intensity plyometric training group and control group and high intensity plyometric training group and control group were 1.60, 2.75, 3.20, 1.15, 4.80, and 5.95, respectively. They were greater than the confidence interval value of 0.788, at .05 level,

which indicates that there was a significant difference among abovesaid six paired groups on leg strength.

Table 2 also shows that the adjusted post test mean difference between low intensity plyometric training group and moderate intensity plyometric training group, low intensity plyometric training group and high intensity plyometric training group, low intensity plyometric training group and control group, moderate intensity plyometric training group and high intensity plyometric training group, moderate intensity plyometric training group and control group and high intensity plyometric training group and

control group were 2.66, 5.67, 5.14, 3.01, 7.80, and 10.81, respectively. They were greater than the confidence interval value of 1.51, at .05 level, which indicates that there was a significant difference among abovesaid six paired groups, on kicking.

The result of the study indicates the low, moderate and high intensity plyometric training had significantly improved on leg strength and kicking. The results of the study are in consonance with the findings of Haddock (2002) who reported significantly improved strength irrespective of training volume (high, low and variety). Hass (1998) noticed low and high volume of strength training increases strength. Fatouros *et al* (2000) concluded that plyometric training improved vertical jump performance and leg strength.

Oberg *et al* (1985) stated that the normal soccer training will increase strength; but an individual strength training programme added to the Soccer training will result in higher strength levels. The result of the study are in consonance

with the findings of (Cabri *et al*, 1987) who confirmed that Soccer players have a better kick performance than non-Soccer players. Better kicking performance could be achieved after strength training of the lower limbs (Manolopoulos *et al*, 2004). These studies are positively related to results of the present study.

CONCLUSION

1. There was a significant improvement on leg strength and kicking performance due to 12 weeks of low, moderate and high intensity plyometric training.
2. There was a significance difference in the improved leg strength and kicking among low intensity plyometric training, moderate intensity plyometric training and high intensity plyometric training.
3. High intensity plyometric training significantly improved leg strength and kicking as compared to low intensity plyometric training and moderate intensity plyometric training.

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